DATA MINING EXAM PREP HANDBOOK

MODULE 1: Introduction & Data Pre-processing

Author: Prof. [Your Name] **Target Audience:** University Students **Tone:** Rigorous, Logical, No-Nonsense

TABLE OF CONTENTS - MODULE 1

1. **Overview:** The Data Explosion & The Need for Mining
2. **The Process:** KDD vs. CRISP-DM
3. **Data Pre-processing (The Core):**
    - Data Cleaning (Handling Missing Values & Noisy Data)
    - Data Integration (Correlation Analysis & Chi-Square)
    - Data Transformation (Normalization Techniques)
    - Data Reduction (Dimensionality & Numerosity)
4. **Exploratory Data Analysis (EDA) & Visualization**
5. **Professor's Corner:** Common Mistakes & Exam Strategy
6. **Question Bank:** MCQs, Short Answer, Long Answer

1. OVERVIEW: THE DATA EXPLOSION

1.1 Definition & Reality Check

Data Mining is not just "analyzing data." It is the automated or semi-automated process of discovering patterns, knowledge, and anomalies in large datasets.

The "Rich Data, Poor Knowledge" Paradox: We are drowning in data but starving for knowledge.

- **Data:** Raw facts (e.g., "Transaction ID 101: Bread, Milk, $5.00").
- **Information:** Aggregated data (e.g., "Bread sales are up 10%").
- **Knowledge:** Actionable patterns (e.g., "Customers who buy milk are 80% likely to buy bread").

1.2 What Data Mining Is NOT

- It is **not** simple SQL querying (`SELECT * FROM sales WHERE price > 100`). That is information retrieval.
- It is **not** just statistics. Statistics verifies hypotheses; Data Mining generates them.

1.3 Real-World Applications (Case Studies)

1. **Market Basket Analysis (Retail):** Walmart discovered the "Beer and Diapers" correlation. On Friday afternoons, young fathers buy diapers and beer. *Action:* Place high-margin beer next to diapers.
2. **Fraud Detection (Finance):** Credit card companies use outlier analysis to detect spending patterns that deviate from a user's historical mean.
3. **Churn Prediction (Telecom):** Identifying customers likely to switch to a competitor based on call drops and usage decline.

2. THE PROCESS: KDD vs. METHODOLOGIES

You must know the **KDD (Knowledge Discovery in Databases)** process by heart. In an exam, if you miss a step, you lose marks.

2.1 The KDD Pipeline

1. **Selection:** Selecting the relevant data from the database.
2. **Pre-processing:** Cleaning and scrubbing the data (removing noise).
3. **Transformation:** Converting data into a format suitable for mining (e.g., normalizing values).
4. **Data Mining:** Applying algorithms (Clustering, Classification, Association) to extract patterns.
5. **Interpretation/Evaluation:** Filtering patterns. Are they interesting? Are they valid?

2.2 Introduction to Data Matrices

Data is usually represented as an $m \times n$ matrix, where m is the number of rows (samples/objects) and n is the number of columns (attributes/features).

$$X = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} & x_{m2} & \cdots & x_{mn} \end{bmatrix}$$

3. DATA PRE-PROCESSING (CRITICAL TOPIC)

Professor's Note: This is where 60-80% of a data scientist's time is spent. In the exam, expect calculation-heavy questions on Normalization and Correlation.

3.1 Data Cleaning

Data in the real world is **Dirty**:

- **Incomplete:** Lacking attribute values, lacking certain attributes of interest, or containing only aggregate data.
- **Noisy:** Containing noise, errors, or outliers.

- **Inconsistent:** Containing discrepancies in codes or names (e.g., "NY" vs. "New York").

Handling Missing Values

1. **Ignore the tuple:** Only useful if the tuple contains several missing values or the dataset is huge.
2. **Fill manually:** Boring and infeasible for large datasets.
3. **Use a global constant:** Fill with "Unknown" or ∞. (Dangerous: model may treat "Unknown" as a real category).
4. **Use the attribute mean:** Fill with the average of the column.
5. **Use the most probable value:** Inference-based (Bayesian or Decision Tree).

Handling Noisy Data (Binning)

Noise is a random error or variance in a measured variable. We smooth it using **Binning**.

Technique: Binning Methods

- **Sorted Data:** 4, 8, 15, 21, 21, 24, 25, 28, 34
- **Partition into 3 bins (Equal-Depth):**
  - Bin 1: 4, 8, 15
  - Bin 2: 21, 21, 24
  - Bin 3: 25, 28, 34

Smoothing Methods:

1. **Smoothing by Bin Means:** Replace all values in the bin with the mean.
   - Bin 1 Mean: $(4 + 8 + 15)/3 = 9$ -> [9, 9, 9]
   - Bin 2 Mean: $(21 + 21 + 24)/3 = 22$ -> [22, 22, 22]
2. **Smoothing by Bin Boundaries:** Replace with the closest boundary (min or max of that bin).
   - Bin 1: 4 is boundary, 15 is boundary. 8 is closer to 4. -> [4, 4, 15]

3.2 Data Integration

Combining data from multiple sources. The main issue is **Redundancy**.

Correlation Analysis (Numerical Data)

We use the **Pearson Correlation Coefficient (r)**.

$$r_{A,B} = \frac{\sum_{i=1}^{N}(a_i - \bar{A})(b_i - \bar{B})}{N\sigma_A\sigma_B}$$

Where:

- N: Number of tuples.

- a_i, b_i: Values of attributes A and B.
- $\bar{A}, \bar{B}$: Mean values.
- σ_A, σ_B: Standard deviations.

Interpretation:

- $r > 0$: Positively correlated (A rises, B rises).
- $r = 0$: Independent.
- $r < 0$: Negatively correlated (A rises, B falls).

WORKED EXAMPLE 1: Correlation Calculation

Dataset:

- A (Hours Studied): 2, 5, 1
- B (Exam Score): 50, 90, 40

Step 1: Calculate Means $\bar{A} = (2+5+1)/3 = 2.66$ $\bar{B} = (50+90+40)/3 = 60$

Step 2: Calculate Standard Deviations (σ) $\sigma_A = \sqrt{\frac{(2-2.66)^2+(5-2.66)^2+(1-2.66)^2}{3}} = 1.69$
$\sigma_B = \sqrt{\frac{(50-60)^2+(90-60)^2+(40-60)^2}{3}} = 21.6$

Step 3: Calculate Covariance (Numerator)
$\text{Cov}(A,B) = \frac{1}{3}[(2-2.66)(50-60) + (5-2.66)(90-60) + (1-2.66)(40-60)]$
$= \frac{1}{3}[(-0.66 \times -10) + (2.34 \times 30) + (-1.66 \times -20)] = \frac{1}{3}[6.6 + 70.2 + 33.2] = 36.66$

Step 4: Pearson r $r = \frac{36.66}{1.69 \times 21.6} \approx 0.99$

Conclusion: Strong positive correlation. Studying more increases scores.

Chi-Square Test (Categorical Data)

Used to determine if two categorical variables are independent.

$$\chi^2 = \sum \frac{(Observed - Expected)^2}{Expected}$$

Expected Value Calculation:

$$E_{ij} = \frac{\text{count}(A_i) \times \text{count}(B_j)}{N}$$

WORKED EXAMPLE 2: Chi-Square

Scenario: Do men and women prefer different genres of books? **Observed Table:** | | Fiction | Non-Fiction | Total | |---|---|---|---| | **Male** | 250 | 200 | 450 | | **Female** | 50 | 1000 | 1050 | | **Total** | 300 | 1200 | 1500 |

Step 1: Calculate Expected Values

- $E(\text{Male, Fiction}) = \frac{450 \times 300}{1500} = 90$
- $E(\text{Male, Non-Fiction}) = \frac{450 \times 1200}{1500} = 360$
- $E(\text{Female, Fiction}) = \frac{1050 \times 300}{1500} = 210$
- $E(\text{Female, Non-Fiction}) = \frac{1050 \times 1200}{1500} = 840$

Step 2: Calculate χ^2 $\chi^2 = \frac{(250-90)^2}{90} + \frac{(200-360)^2}{360} + \frac{(50-210)^2}{210} + \frac{(1000-840)^2}{840}$
$= 284.4 + 71.1 + 121.9 + 30.4 = 507.8$

Conclusion: The value is huge. If the critical value (from a table) is typically around 3.84 for 1 degree of freedom, we reject the null hypothesis. Gender and Genre are **strongly correlated**.

3.3 Data Transformation (Normalization)

Neural Networks and Distance-based algorithms (KNN, K-Means) fail if features are on different scales (e.g., Age: 0-100 vs. Salary: 20k-200k). Normalization solves this.

1. Min-Max Normalization

Scales data to a fixed range, usually [0, 1].

$$v' = \frac{v - \min_A}{\max_A - \min_A}(new_max_A - new_min_A) + new_min_A$$

2. Z-Score Normalization (Zero-Mean)

Useful when outliers exist or min/max are unknown.

$$v' = \frac{v - \bar{A}}{\sigma_A}$$

3. Decimal Scaling

Moves the decimal point.

$$v' = \frac{v}{10^j} \quad (\text{where } j \text{ is the smallest integer such that } \max(|v'|) < 1)$$

WORKED EXAMPLE 3: Normalization

Data: Age = 35. Range of Age = [20, 70]. Mean = 45, SD = 12.

Q1: Min-Max Normalize to [0, 1] $v' = \frac{35-20}{70-20}(1-0) + 0 = \frac{15}{50} = 0.3$

Q2: Z-Score Normalize $v' = \frac{35-45}{12} = \frac{-10}{12} = -0.833$

Q3: Decimal Scaling Max value is 70. To make it < 1, we divide by 100 ($j = 2$).
$v' = 35/100 = 0.35$

4. EXPLORATORY DATA ANALYSIS (EDA)

You must know how to visualize data distribution.

1. **Boxplot:** Displays the 5-number summary:
    - Minimum
    - Q1 (25th percentile)
    - Median (50th percentile)
    - Q3 (75th percentile)
    - Maximum
    - **IQR (Interquartile Range):** $Q3 - Q1$. Outliers are points beyond $1.5 \times IQR$.
2. **Histogram:** Shows frequency distribution.
3. **Scatter Plot:** Shows correlation between two variables.

5. PROFESSOR'S CORNER: EXAM STRATEGY

Common Mistakes

1. **Confusing Mean and Median:** In skewed data (e.g., salaries), Median is better. Don't use Mean blindly.
2. **Forgetting to Normalize:** If an exam question asks for K-Means clustering steps, and you don't normalize the data first, I will deduct 50% of the marks. Distance calculations are meaningless without normalization.
3. **Binning Errors:** When smoothing by bin means, students often calculate the mean but forget to replace *every* value in the bin with that mean.

Tips for High Scores

- **Write Algorithms:** Don't just explain "K-Means" in paragraphs. Write pseudocode steps.
- **Draw Diagrams:** For KDD process, draw the blocks. For Boxplots, label the whiskers and IQR.
- **Show Workings:** In normalization problems, show the formula, then substitution, then the final answer.

6. MODULE 1 QUESTION BANK

Section A: Multiple Choice (1 Mark Each)

1. **Which of the following is NOT a data preprocessing step?** a) Data Cleaning b) Data Visualization c) Data Transformation d) Data Reduction *Answer: b (Visualization is part of EDA, typically considered separate from the core preprocessing pipeline in strict KDD definitions, though often related).*
2. **In Min-Max normalization, if the min is 50, max is 100, and value is 75, what is the normalized value in range [0,1]?** a) 0.25 b) 0.50 c) 0.75 d) 1.0 *Answer: b*

3. **Which technique is used to handle noisy data?** a) Regression b) Clustering c) Binning d) All of the above *Answer: d*

Section B: Short Answer (5 Marks)

1. **Differentiate between Data Characterization and Data Discrimination.**
   - *Hint:* Characterization summarizes the target class (e.g., "What does a Big Spender look like?"). Discrimination compares the target class to a contrasting class (e.g., "Big Spenders vs. Budget Shoppers").
2. **Explain the "Five Number Summary" and how it relates to a Boxplot.**
   - *Hint:* List Min, Q1, Median, Q3, Max. Draw a boxplot and label these 5 points. Explain IQR.

Section C: Long Answer / Numerical (10 Marks)

Q1. Consider the following data for attribute "Age": `13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30`

(a) Partition the data into 3 bins using **Equal-Frequency (Equal-Depth)** partitioning. (b) Smooth the data using **Smoothing by Bin Means**. (c) Smooth the data using **Smoothing by Bin Boundaries**.

Solution Key: Total items = 15. Bins = 3. Items per bin = 5.

(a) Partitions: Bin 1: 13, 15, 16, 16, 19 Bin 2: 20, 20, 21, 22, 22 Bin 3: 25, 25, 25, 25, 30

(b) Bin Means: Bin 1 Mean = 15.8 -> [15.8, 15.8, 15.8, 15.8, 15.8] Bin 2 Mean = 21 -> [21, 21, 21, 21, 21] Bin 3 Mean = 26 -> [26, 26, 26, 26, 26]

(c) Bin Boundaries: Bin 1: (min 13, max 19). 15 is closer to 13. 16 is closer to 19. -> [13, 13, 19, 19, 19] ... (continue for Bins 2 and 3).

Module 2: Data Warehousing & OLAP

Comprehensive Exam-Prep Handbook

Module Duration: 12 Hours **Unit Weight:** High (Core Concept)

2.1 Introduction to Data Warehouse

2.1.1 Definition & Core Concept

A **Data Warehouse (DW)** is a specialized database optimized for analyzing and reporting data rather than processing transactions. Unlike standard databases that record daily interactions (like buying a coffee), a DW aggregates data from multiple sources to support decision-making.

The "Inmon" Definition:

> "A data warehouse is a **subject-oriented**, **integrated**, **time-variant**, and **non-volatile** collection of data in support of management's decision-making process." — *W.H. Inmon*

The "Kimball" Definition:

> "A copy of transaction data specifically structured for query and analysis." — *Ralph Kimball*

2.1.2 The "Why": Historical Context

Before DWs (pre-1990s), organizations relied on operational systems for reports. If a CEO wanted "Total Sales by Region for the last 5 years," the IT team had to run a heavy query on the live database, slowing down checkout lines and crashing systems. The Data Warehouse was born to separate **Workload A (Transactions)** from **Workload B (Analysis)**.

2.2 OLTP vs. OLAP: The Crucial Distinction

This is the most frequent exam topic. Do not confuse them.

Feature	OLTP (Operational/Transactional)	OLAP (Data Warehouse)
Full Form	Online Transaction Processing	Online Analytical Processing
Primary User	Clerks, DBAs, Client Applications	Managers, Executives, Analysts
Focus	Data **IN** (Insert, Update, Delete)	Data **OUT** (Select, Aggregate)
Data Content	Current, real-time data	Historical, archived data

Design Style	ER Model, Normalized (3NF) to avoid redundancy	Star/Snowflake Schema, Denormalized for speed
Access Pattern	Short, atomic read/writes	Long, complex aggregations
Size	Megabytes to Gigabytes	Terabytes to Petabytes
Metric for Success	Transaction throughput (TPS)	Query response time

Exam Tip: If asked to differentiate, do not just list features. Explain the *consequence* of the design.

- *Example:* "OLTP uses normalization to ensure data integrity during updates. OLAP uses denormalization to reduce join complexity during queries."

2.3 Data Warehouse Characteristics (The AC-ID of DW)

You must memorize the four characteristics defined by Inmon.

1. **Subject-Oriented:**
   - *Theory:* Organized around major subjects (Customer, Product, Sales) rather than applications (Order Processing App, Payroll App).
   - *Example:* An operational system has an "Enrollment" table. The DW has a "Student" dimension that combines enrollment, library, and canteen data.
2. **Integrated:**
   - *Theory:* Data from disparate sources (SQL, Excel, Legacy Mainframes) must be made consistent.
   - *Real-World:* Source A uses `sex: m/f`. Source B uses `gender: 1/0`. The DW standardizes this to `gender: Male/Female`.
3. **Time-Variant:**
   - *Theory:* Every data point is associated with a specific point in time. Historical data is kept, not overwritten.
   - *Contrast:* In OLTP, if a customer moves, their address is updated. In DW, the old address is kept (to analyze sales in the old region) and a new record is created.
4. **Non-Volatile:**
   - *Theory:* Once data enters the DW, it is (mostly) read-only. We do not update individual records; we load new batches.

2.4 Data Warehouse Architecture

2.4.1 Three-Tier Architecture

This is the standard architectural model you will be asked to draw.

1. **Bottom Tier (Warehouse Server):**
    - The physical storage (Relational Database).
    - Fed by the ETL tools.
    - Contains the **Meta Data** (data about data) and the **Data Marts**.
2. **Middle Tier (OLAP Server):**
    - The engine that converts relational data into multidimensional structures (Cubes).
    - Can be ROLAP (maps to relational) or MOLAP (direct array storage).
3. **Top Tier (Front-End Tools):**
    - Reporting tools (Tableau, PowerBI), Query tools, Data Mining tools.

2.4.2 ETL: Extraction, Transformation, Loading

The ETL process is the "plumbing" of the data warehouse. It consumes 70-80% of project time.

1. **Extraction:** Reading data from source systems (databases, logs, APIs).
2. **Transformation:**
    - **Cleaning:** Removing noise (e.g., age = 999).
    - **Standardization:** "NY", "N. York", "New York" $\rightarrow$ "NY".
    - **Deduplication:** Merging duplicate customer records.
    - **Calculation:** Deriving "Profit = Sales - Cost".
3. **Loading:** Writing the clean data into the warehouse.
    - *Initial Load:* Populating the empty DW.
    - *Incremental Load:* Adding only new/changed data (e.g., every night).

2.5 Logical Data Modelling: Multidimensional Model

In a DW, we do not use the standard Entity-Relationship (ER) model. We use **Dimensional Modelling**.

2.5.1 The Core Concepts

- **Fact Table:** Contains quantitative data (Measures). These are the numbers you want to analyze.
    - *Examples:* Sales Amount, Quantity Sold, Total Calls, Grade Points.
    - *Structure:* Contains Foreign Keys to dimensions + Measure columns.
- **Dimension Table:** Contains descriptive data (Context). These describe the "Who, What, Where, When".
    - *Examples:* Customer Name, Product Category, Store Location, Month/Year.

2.5.2 Worked Example: Retail Sales

- **Fact:** `Sales_Fact` (Keys: `Date_ID`, `Product_ID`, `Store_ID`. Measures: `Units_Sold`, `Revenue`).
- **Dimension:** `Product_Dim` (Attributes: `Product_Name`, `Category`, `Brand`, `Weight`).
- **Dimension:** `Time_Dim` (Attributes: `Date`, `Month`, `Quarter`, `Year`, `Holiday_Flag`).

2.6 Schema Design: Star, Snowflake, and Constellation

2.6.1 Star Schema

The simplest and most common schema.

- **Structure:** One large central Fact table surrounded by Dimension tables.
- **Visual:** Looks like a star.
- **Normalization:** Dimension tables are **denormalized**.
  - *Example:* In `Location_Dim`, we store City, State, and Country in the same row. We do *not* split them into separate tables.
- **Pros:** Fast queries (fewer joins), simple for users to understand.
- **Cons:** Data redundancy in dimensions (minor issue given cheap storage).

2.6.2 Snowflake Schema

A variation of the Star schema where dimensions are normalized.

- **Structure:** Dimension tables are split into sub-dimensions.
  - *Example:* `Location_Dim` is split into `City_Table`, `State_Table`, `Country_Table`.
- **Pros:** Saves space, easier to maintain standard 3NF structures.
- **Cons:** Complex queries (many joins), slower performance.
- **Exam Strategy: Avoid recommending Snowflake** unless storage space is severely constrained (rare in modern times). Star is usually better.

2.6.3 Fact Constellation (Galaxy Schema)

- **Structure:** Multiple Fact tables sharing common Dimension tables.
- **Use Case:** Enterprise-wide analysis.
  - *Example:* `Sales_Fact` and `Shipping_Fact` both share the `Time_Dim` and `Product_Dim`.

2.7 OLAP Cubes and Operations

2.7.1 The OLAP Cube

A data cube allows data to be modeled and viewed in multiple dimensions.

- **0-D:** A scalar value (Total Sales).
- **1-D:** A list (Sales by Product).

- **2-D:** A table/matrix (Sales by Product and Region).
- **3-D:** A cube (Sales by Product, Region, and Time).
- **N-D:** A hypercube (adding Customer type, Promotion, etc.).

2.7.2 OLAP Operations (The "Big 5")

You must be able to describe these with examples.

1. **Roll-up (Drill-up):**
   - *Action:* Climbing up the hierarchy or reducing dimensions.
   - *Logic:* Summarizing data.
   - *Example:* Moving from "Daily Sales" to "Monthly Sales". Moving from "City" to "Country".
2. **Drill-down:**
   - *Action:* Stepping down the hierarchy or adding dimensions.
   - *Logic:* Getting more detail.
   - *Example:* Expanding "Q1 Sales" to see "Jan, Feb, Mar".
3. **Slice:**
   - *Action:* Selecting a single value for one dimension.
   - *Logic:* Cutting a slice out of the cube.
   - *Example:* Showing all sales for ONLY "Electronics" (fixing the Product dimension).
4. **Dice:**
   - *Action:* Selecting a sub-cube by specifying ranges on two or more dimensions.
   - *Logic:* Selecting a smaller block.
   - *Example:* Sales for "Electronics" AND "Furniture", in "Q1" AND "Q2", for "North Region".
5. **Pivot (Rotate):**
   - *Action:* Rotating the axes of the data view.
   - *Logic:* Changing the perspective.
   - *Example:* Moving "Region" from rows to columns.

2.8 OLAP Server Architectures: ROLAP, MOLAP, HOLAP

Feature	ROLAP (Relational)	MOLAP (Multidimensional)	HOLAP (Hybrid)
Storage	Standard Relational DB (RDBMS)	Optimized Multi-dimensional Arrays	Combination
Speed	Slower (requires SQL joins)	Fastest (pre-calculated)	Fast for summaries, slow for detail

Scalability	High (can handle huge data)	Limited (Cube explosion)	High
Data Latency	Real-time possible	Requires processing time to build cube	Mixed
Use Case	Large data, infrequent queries	Specific analysis, frequent queries	Best of both worlds

2.9 Worked Examples

Example 1: Schema Design for a Hospital

Scenario: A hospital wants to analyze patient admissions. **Step** 1: Identify **Grain.** One row per admission. **Step 2: Identify Fact.** `Admission_Fact`

- Measures: `Duration_Days` , `Bill_Amount` , `Medicine_Count` . **Step 3: Identify Dimensions.**
- `Patient_Dim` (Name, Age, Gender, Blood Type)
- `Doctor_Dim` (Name, Specialization, Shift)
- `Time_Dim` (Date of Admission, Month, Year)
- `Diagnosis_Dim` (ICD Code, Disease Name, Category) **Step 4: Schema.** Connect Fact table to all Dimensions using Foreign Keys.

Example 2: OLAP Operation Tracing

Given: Total Sales = $100k **Op 1 (Drill-down on Time):**

- Q1: $20k, Q2: $30k, Q3: $40k, Q4: $10k. **Op 2 (Slice on Region="North"):**
- We only look at North data. Q1 North: $5k, Q2 North: $10k... **Op 3 (Roll-up on Product):**
- Instead of "iPad" and "iPhone" sales, we sum them up to "Apple Products".

2.10 Exam Strategies & Practice

Common Mistakes to Avoid

1. **Confusing OLTP and OLAP:** This guarantees a loss of marks. Use the "User" and "Volatility" differences to distinguish them.
2. **Star vs. Snowflake:** Do not say Snowflake is "better" because it is normalized. In DW, normalization is often bad for performance. Star is usually the preferred answer.
3. **ETL Order:** It is Extract-Transform-Load. Do not write Extract-Load-Transform (ELT) unless specifically asked about modern Data Lakes.
4. **Mixing** up **Slice and Dice:** Slice = 1 dimension cut. Dice = 2+ dimensions cut.

Practice Questions

Short Answer Questions (2-3 Marks)

1. **Q:** Define "Data Granularity" in the context of a Fact Table.
    - **Hint:** Refers to the level of detail. High grain = atomic data (every scan). Low grain = summarized data (monthly totals).
2. **Q:** Differentiate between a Data Mart and a Data Warehouse.
    - **Hint:** Scope. Warehouse = Enterprise-wide. Mart = Department-specific (Marketing Mart).
3. **Q:** Why is "Time" a critical dimension in every Data Warehouse?
    - **Hint:** DWs are historical. Every fact must be anchored to a point in history to allow trend analysis.

Long Answer Questions (5-10 Marks)

1. **Q:** Explain the Star Schema with a diagram. Discuss its advantages and disadvantages compared to the Snowflake Schema.
    - **Strategy:** Draw a central Fact table with 4 surrounding Dimension tables. Draw lines. Explicitly label primary and foreign keys. List "Simplicity" and "Speed" as advantages.
2. **Q:** Describe the ETL process. What are the common challenges faced during the "Transformation" phase?
    - **Strategy:** Focus on data quality issues: missing values, inconsistent formats (US vs UK dates), and mismatched keys.

Model Answer: Designing a Fact Table

Question: Design a Fact Constellation schema for a Shipping company that tracks `Orders` and `Shipments`. **Answer:**

- **Shared Dimensions:** `Time_Dim`, `Customer_Dim`, `Item_Dim`.
- **Fact Table 1: Order_Fact**
    - Keys: `Time_ID`, `Customer_ID`, `Item_ID`.
    - Measures: `Order_Quantity`, `Order_Amount`.
- **Fact Table 2: Shipment_Fact**
    - Keys: `Time_ID`, `Customer_ID`, `Item_ID`, `Shipper_ID`.
    - Measures: `Shipping_Cost`, `Days_Taken`, `Weight`.
- **Diagram:** Draw two boxes (Facts) in the middle, sharing the common dimension boxes on the outside.

2.11 Unit Cheat Sheet

Term	Definition	Key Keyword

Data Warehouse	Subject-oriented, Integrated, Time-variant, Non-volatile collection.	Decision Making
OLTP	Operational systems for daily transactions.	Current Data
OLAP	Analytical systems for historical reporting.	Historical Data
Fact Table	Central table containing numbers (measures).	Quantitative
Dimension Table	Tables containing descriptors (text).	Descriptive
Star Schema	Denormalized dimensions; 1 join away from fact.	Simple/Fast
Snowflake Schema	Normalized dimensions; many joins.	Space Saving
ETL	Extract, Transform, Load.	Cleaning
Data Mart	A subset of a DW for a specific department.	Departmental
Slice	Selection on one dimension.	Single Cut
Dice	Selection on multiple dimensions.	Sub-cube
Roll-up	Summarize data (climb hierarchy).	Generalize
Drill-down	Expand data (descend hierarchy).	Detail

Mini-Case Study: The "Dirty Data" Nightmare

Context: A bank tried to build a DW but failed. **Problem:** The Source System A recorded Gender as 'M/F', Source System B recorded it as '0/1', Source System C had it as 'Male/Female'. **Result:**

Module 3: Classification and Regression

Comprehensive Exam-Prep Handbook

1. Unit Overview & Historical Context

Classification and **Regression** are the twin pillars of supervised learning. While they share the goal of mapping input variables (X) to an output variable (Y), they differ fundamentally in the nature of Y.

- **Classification:** Y is categorical (discrete). Example: Is this email Spam or Ham?
- **Regression:** Y is continuous (numerical). Example: What will the house price be next year?

Historical Context

- **Regression:** The term "regression" was coined by **Francis Galton** in the 19th century while studying the heights of fathers and sons. He observed "regression to the mean"—tall fathers tended to have sons shorter than themselves, and vice versa.
- **Bayes Theorem:** Named after **Reverend Thomas Bayes** (1701–1761), whose work on probability was published posthumously. It wasn't until the advent of modern computing that Bayesian methods became computationally feasible for large datasets.
- **Decision Trees:** The **CART** (Classification and Regression Trees) algorithm was introduced by Breiman et al. in 1984, revolutionizing interpretable machine learning.

2. Decision Tree Algorithms

A Decision Tree is a flowchart-like structure where:

- **Internal Node:** Represents a test on an attribute (e.g., "Is Age > 30?").
- **Branch:** Represents the outcome of the test.
- **Leaf Node:** Represents a class label or continuous value.

2.1. The Core Algorithms: C4.5 vs. CART

Feature	ID3 / C4.5 (Quinlan)	CART (Breiman)
Criterion	Information Gain / Gain Ratio	Gini Impurity
Splits	Multi-way splits (categorical)	Always Binary splits
Task	Primarily Classification	Classification & Regression
Pruning	Error-based pruning	Cost-complexity pruning

2.2. Splitting Criteria (The Math)

A. Entropy (Used in ID3/C4.5) Entropy measures impurity or disorder. A set is "pure" (Entropy = 0) if all samples belong to the same class.

$$H(S) = -\sum_{i=1}^{c} p_i \log_2(p_i)$$

Where p_i is the probability of class i.

Information Gain (IG): Measures the reduction in entropy after splitting on attribute A.

$$IG(S, A) = H(S) - \sum_{v \in Values(A)} \frac{|S_v|}{|S|} H(S_v)$$

B. Gini Impurity (Used in CART) Measures the probability of misclassifying a randomly chosen element.

$$Gini(S) = 1 - \sum_{i=1}^{c} (p_i)^2$$

Note: Gini is computationally faster than Entropy because it avoids logarithmic calculations.

2.3. Worked Example: Calculating Information Gain

Scenario: We want to predict if a user buys a computer based on "Age". Dataset S (14 entries): 9 Buys = Yes, 5 Buys = No.

Step 1: Calculate Entropy of the whole system $H(S)$

$$P(Yes) = 9/14, P(No) = 5/14$$

$$H(S) = -\frac{9}{14}\log_2(\frac{9}{14}) - \frac{5}{14}\log_2(\frac{5}{14}) \approx 0.940$$

Step 2: Calculate Entropy for Attribute "Age" Assume Age splits into:

- **Youth (5 cases):** 2 Yes, 3 No

$$H(Youth) = -\frac{2}{5}\log_2(0.4) - \frac{3}{5}\log_2(0.6) = 0.971$$

- **Middle-aged (4 cases):** 4 Yes, 0 No

$$H(Middle) = -\frac{4}{4}\log_2(1) - 0 = 0$$

  (Pure set!)

- **Senior (5 cases):** 3 Yes, 2 No

$$H(Senior) = -\frac{3}{5}\log_2(0.6) - \frac{2}{5}\log_2(0.4) = 0.971$$

Step 3: Calculate Weighted Average Entropy

$$H_{Age}(S) = \frac{5}{14}(0.971) + \frac{4}{14}(0) + \frac{5}{14}(0.971) \approx 0.693$$

Step 4: Information Gain

$$IG(S, Age) = H(S) - H_{Age}(S) = 0.940 - 0.693 = 0.247$$

Interpretation: Splitting by "Age" reduces the uncertainty by 0.247 bits.

3. Bayesian Classifiers

3.1. Naive Bayes

Based on the assumption of **class-conditional independence**: features are independent given the class.

$$P(C|X) = \frac{P(X|C)P(C)}{P(X)}$$

Because the denominator $P(X)$ is constant for all classes, we maximize:

$$P(C|X) \propto P(C)\prod_{i=1}^{n} P(x_i|C)$$

Common Mistake (The "Zero Probability" Trap): If an attribute value doesn't appear in the training set for a specific class, the probability becomes 0, wiping out the entire calculation.

- **Solution: Laplace Smoothing**. Add 1 to the numerator and k (number of possible values) to the denominator.

3.2. Bayesian Belief Networks (BBN)

Naive Bayes assumes total independence. BBNs allow us to model *dependencies* using a Directed Acyclic Graph (DAG).

- **Nodes:** Variables.
- **Edges:** Probabilistic dependencies ($X \to Y$ means X influences Y).
- **CPT:** Conditional Probability Table for each node.

4. Evaluation Metrics

4.1. For Classification

Based on the Confusion Matrix:

- **TP:** True Positive
- **TN:** True Negative
- **FP:** False Positive (Type I Error)
- **FN:** False Negative (Type II Error)

Metric	Formula	Use Case
Accuracy	$\frac{TP+TN}{Total}$	Balanced datasets only.
Precision	$\frac{TP}{TP+FP}$	Spam detection (High cost of FP).
Recall (Sensitivity)	$\frac{TP}{TP+FN}$	Cancer diagnosis (High cost of FN).
F1 Score	$2 \cdot \frac{Prec \cdot Rec}{Prec+Rec}$	Harmonic mean; best for imbalanced data.

4.2. For Regression

- **Mean Squared Error (MSE):** $\frac{1}{n}\sum(y_i - \hat{y}_i)^2$. Penalizes large errors heavily.
- **Mean Absolute Error (MAE):** $\frac{1}{n}\sum|y_i - \hat{y}_i|$. More robust to outliers.

5. Ensemble Methods

"The wisdom of crowds." Combining multiple weak learners to create a strong learner.

5.1. Bagging (Bootstrap Aggregating)

- **Concept:** Train multiple models (usually Decision Trees) on different random subsets of the data (with replacement).
- **Aggregation:** Voting (classification) or Averaging (regression).
- **Goal:** Reduces **Variance** (overfitting).
- **Example: Random Forest**.

5.2. Boosting

- **Concept:** Train models sequentially. Each new model focuses on the errors made by the previous ones (re-weighting misclassified instances).
- **Goal:** Reduces **Bias** (underfitting).
- **Example: AdaBoost**, **Gradient Boosting**, **XGBoost**.

5.3. Random Forests (Deep Dive)

Random Forest improves Bagging by adding **feature randomness**. At each split in a tree, the algorithm considers only a random subset of features.

- **Why?** It de-correlates the trees. If one feature is very strong, standard bagging would result in identical trees. Random Forest forces trees to look at other features.

6. Exam Strategies & Common Mistakes

A. The "Overfitting" Trap

- **Mistake:** Students often assume 100% accuracy on training data is good.
- **Reality:** This usually means the model has memorized the noise (Overfitting).
- **Exam Fix:** Always mention "Pruning" for Decision Trees and "Cross-Validation" for general evaluation.

B. Precision vs. Recall Confusion

- **Mnemonic:**
  - **P**recision is about **P**redictions: "Of all the ones I *predicted* positive, how many were right?"
  - **R**ecall is about **R**eality: "Of all the ones that *really are* positive, how many did I find?"

C. Numerical Stability

- In Naive Bayes, multiplying many small probabilities results in **underflow** (computer rounds to zero).
- **Fix:** Work in Log-space. $\log(A \cdot B) = \log(A) + \log(B)$. Sum logs instead of multiplying probabilities.

7. Practice Questions

Section A: Short Answer (5 Marks)

1. Explain the difference between Gini Index and Entropy. Which one is computationally more expensive and why?
2. Why is "Independence" a naive assumption in Naive Bayes? Give a real-world example where this assumption fails.
3. Define Type I and Type II errors. In the context of a fire alarm system, which error is more dangerous?

Section B: Worked Problem (10 Marks)

Problem: Given the following confusion matrix for a customized cancer detection model:

- TP = 90
- FP = 10
- FN = 5
- TN = 895

1. Calculate Accuracy, Precision, Recall, and F1 Score.
2. Critique the model. Is Accuracy a good metric here? Why or why not?

Solution Hint:

- Accuracy = $(90 + 895)/1000 = 0.985$ (98.5%).
- Recall = $90/(90 + 5) = 0.947$.
- Precision = $90/(90 + 10) = 0.90$.
- **Critique:** The dataset is imbalanced (95 positive cases vs 905 negative). Accuracy is misleadingly high. Recall is the critical metric here because missing a cancer diagnosis (FN) is fatal. The Recall of 94.7% is good, but the 5 missed cases are significant.

Section C: Long Answer (15 Marks)

1. Compare and contrast Bagging and Boosting.
    - *Structure your answer:* Definition, Goal (Bias vs Variance), Parallel vs Sequential training, Sensitivity to Outliers, and Examples.
    - *Key Insight:* Bagging is easy to parallelize; Boosting is sequential and harder to parallelize but often achieves higher accuracy.

8. Unit Summary Sheet

Algorithm	Key Formula/Concept	Best For	Weakness
Decision Tree	$IG, Gini$	Interpretability	Prone to overfitting
Naive Bayes	$P(C	X) \propto P(C)\Pi P(x_i	C)$
Random Forest	Bagging + Feature Randomness	High Accuracy, Robust	Black box (hard to interpret)
AdaBoost	Weighted errors	Reducing Bias	Sensitive to noisy data/outliers

Module 4: Clustering and Association Rule Mining

Comprehensive Exam-Prep Handbook

Author: Prof. [Your Name/Persona] **Target Audience:** University Students (CS/Data Science) **Tone:** Rigorous, No-Nonsense, Analytical.

Table of Contents

1. **Unit Overview & Learning Objectives**
2. **Part A: Clustering Algorithms**
    - 2.1 K-Means Clustering: The Iterative Standard
    - 2.2 Hierarchical Clustering: Agglomerative vs. Divisive
    - 2.3 Density-Based Clustering: DBSCAN, OPTICS, Mean Shift
3. **Part B: Association Rule Mining**
    - 3.1 Foundations: Support, Confidence, Lift
    - 3.2 The Apriori Algorithm
    - 3.3 FP-Growth Algorithm
4. **Part C: Evaluation Metrics**
    - 4.1 Internal Validation (Silhouette)
    - 4.2 Rule Evaluation
5. **Part D: Exam Bench**
    - 5.1 "Brutal Truth" Exam Strategies
    - 5.2 Numerical Practice Problems
    - 5.3 Theory & Conceptual Questions
    - 5.4 Cheat Sheet

1. Unit Overview & Learning Objectives

Professor's Note: Stop looking for shortcuts. This module is the mathematical backbone of unsupervised learning. If you think you can "wing it" by just memorizing definitions without understanding the underlying geometry of high-dimensional space or the combinatorial explosion in itemset mining, you are setting yourself up for failure. This chapter requires you to **calculate** by hand and **visualize** algorithms step-by-step.

Key Competencies:

- **Derive** and apply the K-Means objective function.
- **Construct** dendrograms manually using specific linkage criteria.

- **Differentiate** between density-reachability and density-connectivity.
- **Execute** Apriori and FP-Growth traces on transaction datasets.

2. Part A: Clustering Algorithms

Clustering is the task of grouping a set of objects in such a way that objects in the same group (cluster) are more similar to each other than to those in other groups.

2.1 K-Means Clustering

Theory & Math

K-Means is a centroid-based algorithm that minimizes the **Within-Cluster Sum of Squares (WCSS)**, also known as inertia.

Objective Function:

$$J = \sum_{j=1}^{k} \sum_{i=1}^{n} ||x_i^{(j)} - c_j||^2$$

Where:

- k is the number of clusters.
- n is the number of data points.
- c_j is the centroid of cluster j.
- $||x_i^{(j)} - c_j||^2$ is the squared Euclidean distance between point x_i and centroid c_j.

The Algorithm (Expectation-Maximization Style):

1. **Initialization:** Select k random centroids.
2. **Assignment (E-step):** Assign each data point to the nearest centroid based on Euclidean distance.
3. **Update (M-step):** Recalculate centroids by taking the mean of all points assigned to that cluster.
4. **Convergence:** Repeat steps 2-3 until centroids do not change or a tolerance threshold is met.

Worked Example: K-Means Manual Trace

Dataset: $P_1(2,10), P_2(2,5), P_3(8,4), P_4(5,8), P_5(7,5), P_6(6,4), P_7(1,2), P_8(4,9)$. **K=3**.
Initial Centroids: $C_1 = P_1(2,10), C_2 = P_5(7,5), C_3 = P_8(4,9)$.

Iteration 1: Calculate distance from every point to C_1, C_2, C_3.

- $P_1 \to C_1$: Dist=0 (Assign C_1)
- $P_2(2,5) \to C_1$: $\sqrt{(2-2)^2 + (10-5)^2} = 5$
  - $\to C_2$: $\sqrt{(2-7)^2 + (5-5)^2} = 5$

- $\rightarrow C_3$: $\sqrt{(2-4)^2+(5-9)^2} = \sqrt{4+16} \approx 4.47$ (Assign C_3)
- *...Calculation continues for all points...*

New Clusters formed:

- Cluster 1: $\{P_1\}$
- Cluster 2: $\{P_3, P_5, P_6\}$
- Cluster 3: $\{P_2, P_4, P_7, P_8\}$

Update Centroids:

- New $C_1 = (2, 10)$
- New $C_2 = (\frac{8+7+6}{3}, \frac{4+5+4}{3}) = (7, 4.33)$
- New $C_3 = (\frac{2+5+1+4}{4}, \frac{5+8+2+9}{4}) = (3, 6)$

Professor's Warning: In the exam, if you calculate the distance incorrectly in step 1, your entire answer is void. Double-check your arithmetic. I see students lose 10+ marks just because they can't subtract correctly.

Real-World Application

- **Customer Segmentation:** Grouping customers by purchasing behavior (RFM analysis).
- **Image Compression:** Reducing colors in an image by clustering pixel RGB values into k colors.

2.2 Hierarchical Clustering

Unlike K-Means, this method does not require specifying k beforehand. It creates a hierarchy of clusters.

Types

1. **Agglomerative (Bottom-Up):** Start with n clusters (each point is a cluster). Merge the two closest clusters iteratively.
2. **Divisive (Top-Down):** Start with 1 cluster containing all points. Split recursively.

Linkage Methods (Crucial for Exams)

How do we define distance between two *clusters*?

1. **Single Linkage (Min):** Distance between the closest pair of points.
   - $$D(C_i, C_j) = \min_{x \in C_i, y \in C_j} d(x, y)$$
   - *Flaw:* Suffers from the "chaining effect" (long, stringy clusters).
2. **Complete Linkage (Max):** Distance between the farthest pair of points.

- $$D(C_i, C_j) = \max_{x \in C_i, y \in C_j} d(x, y)$$
  - *Result:* Tends to produce compact, spherical clusters.
3. **Average Linkage:** Average distance between all pairs.

Worked Example: Dendrogram Construction

Distance Matrix: | | A | B | C | D | |---|---|---|---|---| | **A** | 0 | 1 | 4 | 5 | | **B** | 1 | 0 | 2 | 6 | | **C** | 4 | 2 | 0 | 3 | | **D** | 5 | 6 | 3 | 0 |

Using Single Linkage:

1. **Find min distance:** $d(A, B) = 1$. Merge $\{A, B\}$.
2. **Update Matrix:** Distance from $\{A, B\}$ to C:
   - $d(\{A, B\}, C) = \min(d(A, C), d(B, C)) = \min(4, 2) = 2$.
   - Distance from $\{A, B\}$ to D: $\min(5, 6) = 5$.
3. **Next min:** $d(\{A, B\}, C) = 2$. Merge $\{A, B, C\}$.
4. **Final merge:** $\{A, B, C\}$ with D at distance 3 ($d(C, D) = 3$).

2.3 Density-Based Clustering

Standard algorithms (K-Means) fail on non-convex shapes (e.g., concentric circles, moons). Density-based methods solve this.

DBSCAN (Density-Based Spatial Clustering of Applications with Noise)

Parameters:

- ϵ (Eps): The radius of the neighborhood.
- MinPts: Minimum points required to form a dense region.

Core Concepts:

1. **Core Point:** Has $\geq$ MinPts within Eps radius.
2. **Border Point:** Has $<$ MinPts within Eps, but is in the neighborhood of a Core Point.
3. **Noise Point:** Neither Core nor Border.

Algorithm Logic: Pick a random unvisited point. If it's a Core point, expand the cluster by adding all density-reachable points. If it's Noise, move on (it might be picked up later as a Border point).

Advantages:

- Does not require k.
- Robust to outliers (noise).
- Can find arbitrary shapes.

OPTICS (Ordering Points To Identify the Clustering Structure)

DBSCAN fails if clusters have *varying densities*. OPTICS overcomes this by not explicitly producing clusters, but ordering points such that spatially closest points become neighbors in the ordering. It produces a **Reachability Plot**.

Mean Shift

A sliding-window algorithm that attempts to find dense areas of data points. It is centroid-based but works by updating candidates for centroids to be the mean of the points within a given region.

- **Analogy:** Hiking up a hill in the fog. You step in the direction of the steepest gradient (highest density of data).

3. Part B: Association Rule Mining

Goal: Discover interesting relations between variables in large databases. **Classic Example:** Beer and Diapers.

3.1 Foundations: The Metrics

For rule $A \Rightarrow B$:

1. **Support:** How often does the itemset appear?

$$\text{Support}(A \Rightarrow B) = \frac{\text{Transactions containing } A \cup B}{\text{Total Transactions}}$$

2. **Confidence:** If A is bought, how likely is B bought?

$$\text{Confidence}(A \Rightarrow B) = \frac{\text{Support}(A \cup B)}{\text{Support}(A)}$$

   - *Critique:* Can be misleading if B is popular generally.

3. **Lift:** Is A actually increasing the probability of B?

$$\text{Lift}(A \Rightarrow B) = \frac{\text{Confidence}(A \Rightarrow B)}{\text{Support}(B)}$$

   - Lift = 1: Independent (No correlation).
   - Lift > 1: Positive correlation (Useful rule).
   - Lift < 1: Negative correlation (Substitutes).

3.2 The Apriori Algorithm

The Apriori Principle: If an itemset is frequent, then all of its subsets must also be frequent. (Anti-monotonicity).

Algorithm Steps:

1. **Join Step:** Generate candidate itemsets of length k (C_k) from frequent itemsets of length $k-1$ (L_{k-1}).
2. **Prune Step:** Scan DB to count support for candidates. Discard those below `min_sup`. Apply Apriori property to remove candidates containing infrequent subsets without scanning DB.

Worked Example: Apriori

Min Support = 50% (2 transactions). **T-DB:** T1: {A, C, D} T2: {B, C, E} T3: {A, B, C, E} T4: {B, E}

Step 1: 1-itemsets

- A: 2 (Pass)
- B: 3 (Pass)
- C: 3 (Pass)
- D: 1 (Fail - Prune)
- E: 3 (Pass)
- $L_1 = \{A, B, C, E\}$

Step 2: Generate C_2 from L_1 Pairs: {AB, AC, AE, BC, BE, CE}

- Scan DB for counts:
  - AB: 1 (Fail)
  - AC: 2 (Pass)
  - AE: 1 (Fail)
  - BC: 2 (Pass)
  - BE: 3 (Pass)
  - CE: 2 (Pass)
- $L_2 = \{AC, BC, BE, CE\}$

Step 3: Generate C_3 from L_2 Join L_2:

- Combine BC and BE $\rightarrow$ BCE.
- Combine BC and CE $\rightarrow$ BCE.
- Check subsets of BCE: BC (freq), BE (freq), CE (freq). Valid candidate.
- Scan DB for {B, C, E}: Occurs in T2, T3. Count = 2.
- $L_3 = \{BCE\}$.

3.3 FP-Growth Algorithm (Frequent Pattern Growth)

Apriori is slow because it scans the database repeatedly. FP-Growth scans the DB only twice.

1. **Pass 1:** Count frequencies, sort items in descending order of support.
2. **Pass 2:** Construct the FP-Tree.

3. **Mining:** Recursively mine the tree using conditional pattern bases.

Advantages:

- No candidate generation.
- Compact data structure.
- Much faster than Apriori for dense datasets.

4. Part C: Evaluation Metrics

4.1 Silhouette Coefficient

Used to validate consistency within clusters of data. For point i:

1. $a(i)$: Average distance to all other points in the *same* cluster (Cohesion).
2. $b(i)$: Average distance to all points in the *nearest* cluster (Separation).
3. $$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$$

Interpretation:

- Range: $[-1, 1]$.
- $+1$: Well clustered.
- 0: Overlapping clusters.
- -1: Wrong cluster assignment.

5. Part D: Exam Bench

5.1 "Brutal Truth" Exam Strategies

1. **Don't describe algorithms with text blocks.** If asked for Apriori, I expect pseudocode or a flow chart. A wall of text gets you a C.
2. **Metric Confusion:** Confusing "Support" and "Confidence" is the most common reason for failing the Association Rule question. Memorize the denominators.
   - Support Denominator = Total Transactions.
   - Confidence Denominator = Support of Antecedent (Left hand side).
3. **Dendrogram Scaling:** When drawing dendrograms, the Y-axis represents *distance*. If your merges aren't at the correct height relative to the distance matrix, you lose precision marks.

5.2 Practice Questions

Section A: Multiple Choice (2 Marks Each)

Q1. Which of the following is NOT required for DBSCAN? A) Epsilon B) MinPts C) Number of Clusters (k) D) Distance Metric *Answer: C. DBSCAN discovers k automatically.*

Q2. If Lift(A->B) = 0.5, what does this imply? A) A and B are independent. B) A and B are positively correlated. C) A and B are negatively correlated (substitutes). D) B is twice as likely as A. *Answer: C.*

Section B: Numerical/Long Answer (10 Marks Each)

Q3. Given the 1D dataset $\{2, 4, 10, 12, 3, 20, 30, 11, 25\}$, perform Hierarchical Clustering using **Single Linkage**. Show the distance matrix updates for the first 3 merges. *Hint: Sort the data first to find nearest neighbors easily in 1D.*

Q4. Compare and Contrast K-Means and DBSCAN. Provide a sketch of a dataset where K-Means would fail but DBSCAN would succeed. *Model Answer Strategy: Draw "Concentric Circles" or "Two Moons". Explain that K-Means seeks convex/spherical shapes based on centroids, while DBSCAN seeks density continuity.*

Q5. A retail database has 1,000,000 transactions.

- Item A appears in 20,000 transactions.
- Item B appears in 10,000 transactions.
- Items A and B appear together in 5,000 transactions. Calculate Support(A->B), Confidence(A->B), and Lift(A->B). Is this a strong rule?
- **Solution:**
  - Sup(A) = 0.02, Sup(B) = 0.01, Sup(AB) = 0.005
  - Rule Support = 0.005 (0.5%)
  - Confidence = 0.005 / 0.02 = 0.25 (25%)
  - Lift = 0.25 / 0.01 = 25.
  - **Analysis:** High Lift (25) indicates a very strong correlation, even though Support is low.

Cheat Sheet: Formulas

Metric	Formula
Euclidean Distance	$\sqrt{\sum(x_i - y_i)^2}$
Manhattan Distance	$\sum
Support(X)	$\frac{\text{count}(X)}{N}$
Confidence(X->Y)	$\frac{\text{support}(X \cup Y)}{\text{support}(X)}$
Lift(X->Y)	$\frac{\text{confidence}(X \to Y)}{\text{support}(Y)}$

Module 5: Advanced Topics in Data Mining

Estimated Study Time: 13 Hours **Context:** This module moves beyond structured, tabular data into unstructured (text), graph-based (social networks), and temporal (time series) data, as well as the infrastructure (Big Data) and constraints (Ethics) required to handle them.

5.1 Text Mining and Sentiment Analysis

1.1 Theoretical Framework

Definition: Text mining (or Text Analytics) is the process of deriving high-quality information from text. It involves the discovery of patterns and trends in unstructured text data.

The "Bag of Words" (BoW) Model: The simplest assumption in text mining is that grammar and word order are irrelevant to the topic. We treat a document as a "bag" containing words.

- **Tokenization:** Breaking text into words (tokens).
- **Stop Word Removal:** Removing common words (the, is, at) that carry little semantic weight.
- **Stemming/Lemmatization:** Reducing words to roots (e.g., "running" $\rightarrow$ "run").

TF-IDF (Term Frequency-Inverse Document Frequency): This is the critical statistical measure used to evaluate how important a word is to a document in a collection (corpus).

- **TF (Term Frequency):** How often term t appears in document d.

$$TF(t, d) = \frac{\text{count of } t \text{ in } d}{\text{total words in } d}$$

- **IDF (Inverse Document Frequency):** Measures how rare a term is across the corpus. If a word is common (like "data"), its IDF is low.

$$IDF(t) = \log\left(\frac{N}{DF(t)}\right)$$

  Where N is total documents and $DF(t)$ is number of documents containing t.

- **TF-IDF Score:**

$$w_{t,d} = TF(t, d) \times IDF(t)$$

Sentiment Analysis (Opinion Mining): Computational study of opinions, sentiments, and emotions expressed in text.

- **Levels:** Document-level (overall tone), Sentence-level, Aspect-based (e.g., liking the *camera* but hating the *battery* of a phone).
- **Approaches:**

1. **Lexicon-based:** Counting positive/negative words using a dictionary (e.g., SentiWordNet).
2. **Machine Learning:** Training classifiers (Naive Bayes, SVM) on labeled datasets.

1.2 Worked Examples

Example 1: Calculating TF-IDF Corpus:

- Doc A: "The sky is blue."
- Doc B: "The sun is bright."

Task: Calculate TF-IDF for the word "blue" in Doc A.

Solution:

1. **Preprocessing:** Remove stop words ("The", "is").
   - Doc A: [sky, blue] (Total words = 2)
   - Doc B: [sun, bright] (Total words = 2)
2. **Calculate TF("blue", Doc A):**
   - Count = 1. Total = 2.
   - $TF = 1/2 = 0.5.$
3. **Calculate IDF("blue"):**
   - $N = 2.$
   - $DF(”blue”) = 1$ (appears only in Doc A).
   - $IDF = \log(2/1) = \log(2) \approx 0.301.$
4. **Calculate TF-IDF:**
   - $0.5 \times 0.301 = 0.1505.$

Example 2: Sentiment Classification (Naive Bayes) Training Data:

- "Great movie" (+1)
- "Bad movie" (-1) **Test Data:** "Great acting"

Reasoning: A standard Naive Bayes classifier calculates $P(Positive|”Great”, ”acting”)$. Since "acting" is unseen, we use Laplace Smoothing to avoid zero probability. "Great" has a high probability given label (+1), shifting the classification to Positive.

1.3 Practice Questions

1. **MCQ:** Which technique is used to reduce "running", "runs", and "ran" to "run"?
   - A) Tokenization
   - B) Stop word removal
   - C) Stemming
   - D) Vectorization

- *Answer: C*

2. **Short Answer:** Explain why IDF is necessary. Why can't we just use Term Frequency?
   - *Hint:* Consider the word "the". It has high frequency but zero informational value. IDF penalizes these common words.
3. **Long Answer:** Describe the architecture of a Sentiment Analysis pipeline for analyzing Twitter data. Discuss challenges specific to tweets (slang, sarcasm, character limit).

5.2 Social Network Analysis (SNA) and Link Analysis

2.1 Theoretical Framework

Definition: SNA maps and measures relationships and flows between people, groups, organizations, computers, or other information/knowledge processing entities.

Graph Basics:

- **Nodes (Vertices):** Entities (e.g., Users).
- **Edges (Links):** Relationships (e.g., Follows, Friends).
- **Directed vs. Undirected:** Facebook friendship is undirected (mutual); Twitter follow is directed (one-way).

Centrality Measures (Who is important?):

1. **Degree Centrality:** Number of connections. "The Celebrity".

$$C_D(v) = \frac{deg(v)}{N-1}$$

2. **Betweenness Centrality:** How often a node acts as a bridge along the shortest path between two other nodes. "The Broker".
3. **Closeness Centrality:** Average length of shortest paths to all other nodes. "The Gossip".

Link Analysis Algorithms:

- **PageRank (Google):** A node is important if other important nodes link to it. It models a "random surfer".

$$PR(A) = (1-d) + d\left(\frac{PR(T_1)}{C(T_1)} + \cdots + \frac{PR(T_n)}{C(T_n)}\right)$$

  Where d is the damping factor (usually 0.85).
- **HITS (Hyperlink-Induced Topic Search):** Distinguishes between **Hubs** (resource lists) and **Authorities** (content creators).
  - **Authority Update:** Sum of Hub scores that point to it.
  - **Hub Update:** Sum of Authority scores it points to.

2.2 Worked Examples

Example 3: Calculating Degree Centrality Graph:

- Node A is connected to B, C, D.
- Node B is connected to A.
- Node C is connected to A.
- Node D is connected to A.
- Total Nodes (N) = 4.

Task: Calculate Normalized Degree Centrality for A and B.

Solution:

1. **Node A:**
   - Degree = 3.
   - Max possible connections = $N - 1 = 3$.
   - $C_D(A) = 3/3 = 1.0$ (Star node).
2. **Node B:**
   - Degree = 1.
   - $C_D(B) = 1/3 = 0.33$.

Example 4: PageRank Iteration Scenario:

- Page A links to Page B.
- Page B links to Page A.
- Damping factor $d = 0.85$.
- Initial PR = 1 for both.

Iteration 1 Calculation for A:

- Incoming link from B. B has 1 outgoing link.
- $PR(A) = (1 - 0.85) + 0.85 \times (PR(B)/1)$
- $PR(A) = 0.15 + 0.85 \times 1 = 1.0$. *(Note: In a closed loop of 2 nodes, values stabilize quickly. In complex graphs, they converge over iterations).*

2.3 Practice Questions

1. **MCQ:** A node that connects two dense clusters but has few connections itself likely has high:
   - A) Degree Centrality
   - B) Betweenness Centrality
   - C) Closeness Centrality
   - D) Eigenvector Centrality
   - *Answer: B*

2. **Short Answer:** Explain the difference between a Hub and an Authority in the HITS algorithm. Can a page be both?

5.3 Time Series Analysis and Forecasting

3.1 Theoretical Framework

Definition: A sequence of data points collected at specific time intervals.

Components of Time Series:

1. **Trend (T):** Long-term movement (e.g., increasing sales over 5 years).
2. **Seasonality (S):** Repeating patterns at fixed intervals (e.g., ice cream sales in summer).
3. **Cyclical (C):** Fluctuations not of fixed period (e.g., economic recessions).
4. **Irregular/Noise (I):** Random residuals.

Smoothing Techniques:

- **Simple Moving Average (SMA):** Average of the last k periods.

$$F_{t+1} = \frac{D_t + D_{t-1} + \cdots + D_{t-k+1}}{k}$$

- **Exponential Smoothing:** Gives more weight to recent data.

$$F_{t+1} = \alpha D_t + (1 - \alpha) F_t$$

  Where α is the smoothing constant ($0 < \alpha < 1$).

ARIMA (AutoRegressive Integrated Moving Average): The gold standard for stationary time series.

- **AR (AutoRegressive):** Uses past values to predict future.
- **I (Integrated):** Differencing the data to make it stationary (removing trend).
- **MA (Moving Average):** Uses past forecast errors.

3.2 Worked Examples

Example 5: Moving Average Data: Sales for Jan (10), Feb (12), Mar (16), Apr (13). **Task:** Forecast May using 3-month SMA.

Solution:

$$F_{May} = \frac{\text{Apr} + \text{Mar} + \text{Feb}}{3}$$

$$F_{May} = \frac{13 + 16 + 12}{3} = \frac{41}{3} = 13.67$$

Example 6: Exponential Smoothing Data:

- Forecast for Feb (F_{Feb}) = 105.
- Actual Sales in Feb (D_{Feb}) = 110.
- Alpha (α) = 0.2.

Task: Forecast for March (F_{Mar}).

Solution:

$$F_{Mar} = \alpha D_{Feb} + (1 - \alpha) F_{Feb}$$

$$F_{Mar} = 0.2(110) + 0.8(105)$$

$$F_{Mar} = 22 + 84 = 106$$

Interpretation: The forecast adjusted slightly upward due to the actual sales being higher than predicted.

5.4 Big Data and Distributed Data Mining

4.1 Theoretical Framework

The 5 Vs of Big Data: Volume, Velocity, Variety, Veracity, Value.

Distributed Computing: Standard data mining algorithms (like K-Means or Apriori) fail on single machines when data exceeds RAM. We must use distributed frameworks.

Hadoop & MapReduce:

- **HDFS (Hadoop Distributed File System):** Splits huge files into blocks (e.g., 128MB) and distributes them across nodes.
- **MapReduce Paradigm:**
  1. **Map Step:** Process local data and emit (key, value) pairs.
  2. **Shuffle/Sort:** Group data by key.
  3. **Reduce Step:** Aggregate results for each key.

Apache Spark: Successor to Hadoop MapReduce. Uses **RDDs (Resilient Distributed Datasets)** to keep data in memory (RAM) rather than writing to disk after every step, making it 100x faster for iterative algorithms like K-Means.

4.2 Worked Example

Example 7: MapReduce Word Count Input: "Deer Bear River", "Car Car River", "Deer Car Bear"

Step 1: Map (Deer, 1), (Bear, 1), (River, 1) (Car, 1), (Car, 1), (River, 1) (Deer, 1), (Car, 1), (Bear, 1)

Step 2: Shuffle (Sort by Key) Bear: [1, 1] Car: [1, 1, 1] Deer: [1, 1] River: [1, 1]

Step 3: Reduce (Sum values) (Bear, 2) (Car, 3) (Deer, 2) (River, 2)

5.5 Ethical Considerations and Privacy Issues

5.1 Theoretical Framework

Data mining reveals hidden patterns, often exposing sensitive personal information.

Key Issues:

1. **Privacy:** Application of mining to sensitive data (medical, financial).
2. **Discrimination:** Algorithmic bias (e.g., loan approval models denying specific demographics).
3. **Transparency:** "Black Box" models (Deep Learning) where decisions cannot be explained.

Privacy-Preserving Techniques:

- **Anonymization:** Removing explicit identifiers (Name, SSN). *Critique: Not enough.*
- **k-Anonymity:** Ensures a record is indistinguishable from at least $k-1$ other records regarding quasi-identifiers (Zip, Age, Gender).
- **l-Diversity:** Ensures that within a k-anonymous group, there are at least l distinct values for sensitive attributes (e.g., Disease).
- **Differential Privacy:** Adding noise to query results so that the presence or absence of a single individual cannot be detected.

5.2 Worked Examples

Example 8: k-Anonymity Failure Dataset: | Age | Zip | Disease | |---|---|---| | 25 | 11000 | HIV | | 25 | 11000 | Flu |

If $k=2$, this group looks anonymous based on Age/Zip. However, if an attacker knows Bob is 25 and lives in 11000, they know Bob is in this group. If *all* entries in the group have "HIV", anonymity is preserved, but privacy is breached (Homogeneity Attack). This requires **l-Diversity**.

5.6 Module Summary & Exam Strategies

Quick Revision Table

Topic	Key Algorithm/Concept	Primary Application
Text Mining	TF-IDF, Stemming	Search Engines, Spam Filters
Sentiment Analysis	Naive Bayes, Lexicons	Brand Monitoring, Review Analysis

SNA	PageRank, Betweenness	Influencer Marketing, Fraud Detection
Time Series	ARIMA, Exponential Smoothing	Stock Market, Demand Forecasting
Big Data	MapReduce, Spark RDD	Web Crawling, Log Analysis
Privacy	k-Anonymity, Differential Privacy	Medical Records, Census Data

Exam Strategies for Module 5

1. **Don't Confuse Graph Metrics:** A common mistake is confusing **Betweenness** (bridge) with **Degree** (popularity). If the question asks about "controlling information flow," calculate Betweenness.
2. **Manual Calculations:** Be prepared to manually calculate one iteration of PageRank or a few steps of TF-IDF. Do not rely on intuition; show the formula.
3. **MapReduce Logic:** In "design a MapReduce algorithm" questions, explicitly write out the input and output of the **Map** function and the **Reduce** function.
4. **Privacy vs. Utility:** In essay questions about ethics, always mention the trade-off: higher privacy (more noise/generalization) usually means lower utility (less accurate mining results).

Model Long Answer Question

Q: "Discuss the challenges of mining data streams compared to static databases."

Model Answer Structure:

- **Definition:** Define data streams (continuous, high velocity, infinite).
- **Challenge 1: One-pass constraint.** You cannot store the data; you must process it once and discard it.
- **Challenge 2: Concept Drift.** The statistical properties of the target variable change over time (e.g., spam patterns change). Static models fail here.
- **Solution:** Mention "Sliding Window" models or "Hoeffding Trees" as solutions.